



Day 10 Notes



Smoothing, Sliders and Mirror View

★ Today = made the shirt look and behave properly. Yesterday's version worked, but it jittered and the fit was a guess.

1 Smoothing the Jitter

Pose landmarks wobble a tiny bit every frame, so the shirt shook. Fix: each frame, move only PART of the way toward the new position.

```
const SMOOTHING = 0.35

function smoothValue(previous, next) {
  return previous + (next - previous) * SMOOTHING
}
```

- 1 = no smoothing (jittery)
- 0.1 = very smooth (but laggy)
- 0.35 was a good middle for me

I smooth every value: x, y, width, torso length, and angle.

2 Live Sliders for Tuning

Instead of guessing the multipliers (2, 1.4, 0.12), I added 3 sliders and tuned against my own body:

- Shirt width → how wide, compared to shoulder width
- Shirt height → how long, compared to torso length
- Move shirt up/down → shifts it relative to the shoulders

I wrote down my final slider values, because they become the defaults in the real app.

3 Hiding the Shirt When I Leave

```
function isVisible(p) {
  return (p.visibility ?? 1) > 0.5
}

if (!shouldersVisible) {
  smoothed = null    // forget the old position
  ...return
}
```

Every landmark has a visibility score (0 to 1) saying how sure the model is that the point is really seen. Shoulders not visible = no shirt. Setting smoothed to null means the shirt does NOT slide in from the old spot when I come back.

4 *When My Hips Aren't Visible*

Laptop webcams mostly show the upper body. So instead of hiding the shirt, I estimate:

```
let torsoLength = shoulderWidth * 1.5
if (hips are visible) {
  torsoLength = real distance to hips
}
```

A fallback = the feature still works with a worse view, instead of breaking.

5 *Mirror View*

People expect a try-on to act like a mirror. I flipped ONLY the video, not the canvas, so the shirt itself is never reversed:

```
<video ... style="transform: scaleX(-1);">

function toPixels(p) {
  return { x: (1 - p.x) * canvas.width, y: p.y * canvas.height }
}
```

- `scaleX(-1)` (CSS) → mirrors the video picture
- `(1 - p.x)` → flips the landmark x to match the mirrored video
- After flipping, the angle goes left shoulder to right shoulder on screen

Big takeaway:

Smoothing trades steadiness against lag. Sliders let me tune by eye instead of guessing. Visibility scores tell me when a body part is really seen. Fallbacks keep the feature working with a bad view. Next (Day 11): move this into the real project as a React component, on my own branch feature/tryon, using AntigraVity.

